# Ghifar Khder

Biomedical Engineer | M.Sc. Medical Systems Engineering student

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Portfolio | GitHub | LinkedIn



## Profile

Biomedical engineering graduate studying Medical Systems Engineering at OVGU Magdeburg. My project work includes medical image analysis, machine learning and biomedical electronics. I enjoy understanding how a system works, building it and testing it. I am interested in medical device development, particularly imaging and MRI technology.

## Education

**M.Sc. Medical Systems Engineering** **Apr 2026 - present**

Otto von Guericke University Magdeburg, Germany

Faculty of Electrical Engineering and Information Technology.

**B.Sc. Biomedical Engineering** **Sep 2018 - Aug 2023**

Al Andalus University for Medical Sciences, Syria

Final grade: German equivalent 2.0. Ranked among the top 5 students in the graduating class.

## Projects

### Hardware projects · View in portfolio

**Automatic CPR compression prototype**

- DC motor and rack-and-pinion mechanism with selectable 4, 5 and 6 cm travel and adjustable compression rate.
- The entire design uses no microcontroller. Discrete control circuitry handles motor reversal and position sensing. An RC filter and NE555 pulse shaper address switch bounce at the endpoints.

**Wearable pulse oximeter**

- Wrist-worn prototype combining a MAX30100 sensor, ESP32, OLED display and rechargeable battery to display pulse rate and $SpO_2$.
- Includes browser-based monitoring over Wi-Fi, a rolling measurement log and CSV export. Compared prototype readings with a commercial fingertip oximeter.

**Experimental optical glucose sensing prototype**

- 940 nm infrared emitter and phototransistor with an LM358 amplifier, low-pass filter and Arduino readout.
- Explored a voltage-to-glucose calibration using paired home-meter readings.

### Software projects · View in portfolio

**Knee osteoarthritis grading**

- Built a ResNet classifier for five knee osteoarthritis grades, using a conditional GAN to balance training data.
- Deployed a Streamlit app with image upload, grade prediction and Grad-CAM heatmaps. GitHub

**Coronary stenosis detection**

- Used ResNet101V2 to localise stenosis in about 8,300 coronary angiography images. Evaluated bounding boxes using IoU metrics and deployed a Streamlit app. GitHub

**Diabetes prediction**

- Combined five machine learning models with an ANN meta-learner on the Pima Indians Diabetes dataset. Included missing-value estimation and fuzzy clustering.
- Built an interface for records and CSV files, with model comparisons and predictions. GitHub

**EEG sleep phase classification**

- Trained a 1D CNN to classify two-second, single-channel EEG segments into CAP phases A and B. Built an interface to plot signals and compare predictions with reference labels. GitHub

**Crop disease and yield prediction** · GitHub

## Experience

**Founder and Team Leader** **Jan 2022 - Apr 2026**

On Time Team | Volunteer | Latakia, Syria

- Founded and coordinated a team supporting students with technical projects. Organised work, reviewed progress and acted as the main student contact.
- Designed hardware with the team and completed most programming projects independently.

**Laboratory Receptionist and Workflow Coordinator** **Jan 2023 - Mar 2024**

Private laboratory | Latakia, Syria

- Managed patient records, enquiries, results and payments; coordinated with laboratory staff.

**Department Lead** **Jan 2022 - Jan 2023**

Mad Matrix Team | Volunteer | Tartous, Syria

- Collected students' project ideas, discussed and planned them with students and team members, connected people and contributed to each stage of the team's projects.

## Skills

**Personal skills:** Strong analytical thinking, a solid understanding of mathematics, and the ability to learn new concepts and tools very quickly.

**Data processing and machine learning:** Strong knowledge of data and image processing and analysis; deep learning with ANNs, CNNs and residual CNNs, machine learning and clustering algorithms, transfer learning, ensemble methods, and model evaluation principles.

**Programming and tools:** Python, GitHub, Streamlit, Microsoft Office.

**Electronics:** Circuit design, analysis and assembly, electrical measurements, analog amplification and filtering, sensor integration. Experience with Arduino and ESP32, and the ability to learn other microcontroller platforms very quickly.

## Languages

Arabic: native | English: C1 | German: B1-B2